DATA PREPARATION II

SESSION 5
COMP6140001 – DATA MINING

SUBJECT MATTER EXPERT
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LEARNING OBJECTIVE

LO1 : Explain concept of data, data warehouse, data preprocessing, and data mining

OUTLINE

- Data Transformation
- Dimensionality Reduction

DATA TRANSFORMATION

DATA TRANSFORMATION

- A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values
- Methods
  - Smoothing: remove noise from data
  - Attribute/feature construction: new attributes constructed from the given ones
  - Aggregation: summarization, data cube construction
  - Normalization: scaled to fall within a smaller, specified range
  - Discretization: concept hierarchy climbing

DATA TRANSFORMATION : NORMALIZATION

Normalization

- Min-max normalization: to $[new\_min_A, new\_max_A]$

$$\boxed{v' = \frac{v - min_A}{max_A - min_A}(new\_max_A - new\_min_A) + new\_min_A}$$

- Z-score normalization (μ: $mean$, σ: $standard\ deviation$):

$$\boxed{v' = \frac{v - \mu_A}{\sigma_A}}$$

- Normalization by decimal scaling:

$$\boxed{v' = \frac{v}{10^j}}$$

where j is the smallest integer such that $Max(|v'|) < 1$.

DATA TRANSFORMATION : EXAMPLE NORMALIZATION

- Example:

  E.g., let income range \$12,000 to \$98,000 normalized to [0.0, 1.0] while μ=54,000 and σ=16,000. What is \$73,000 mapped to?

  Answer:

  - <u>Min-max normalization</u>

    $$\frac{73{,}000 - 12{,}000}{98{,}000 - 12{,}000}(1.0 - 0.0) + 0.0 = 0.7093$$

  - <u>Z-score normalization</u>

    $$\frac{73{,}000 - 54{,}000}{16{,}000} = 1.1875$$

  - <u>Normalization by decimal scaling</u>

    $$\frac{73{,}000}{10^5} = 0.73$$

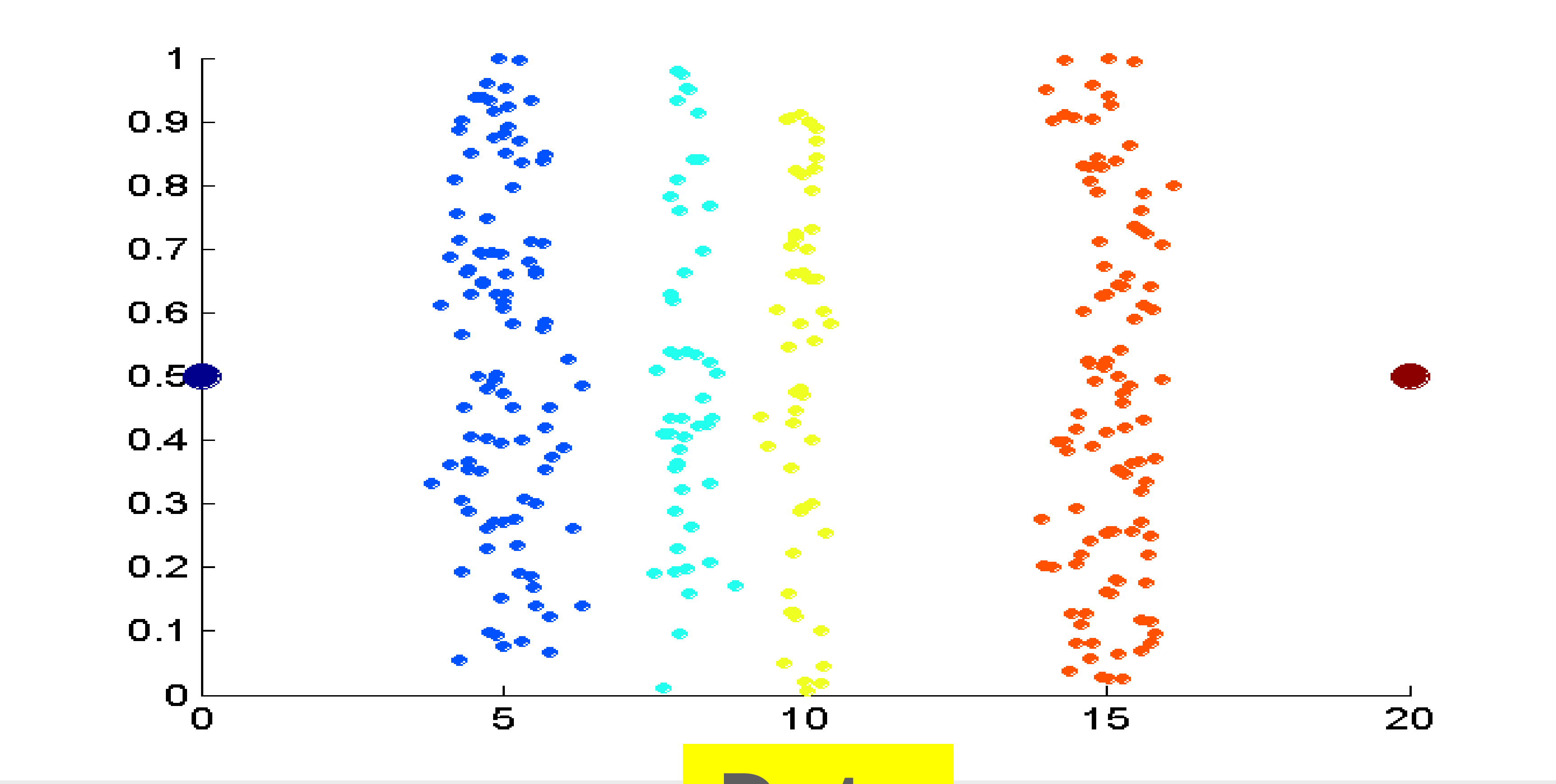
DATA TRANSFORMATION : DISCRETIZATION

- **Discretization:** Divide the range of a continuous attribute into intervals
    - Interval labels can then be used to replace actual data values
    - Reduce data size by discretization
    - Supervised vs. unsupervised
    - Split (top-down) vs. merge (bottom-up)
    - Discretization can be performed recursively on an attribute
    - Prepare for further analysis, e.g., classification
- All **methods** can be applied recursively, i.e., <u>binning</u>, <u>histogram</u>, <u>clustering</u>, <u>decision tree</u>, and <u>correlation analysis</u>, …

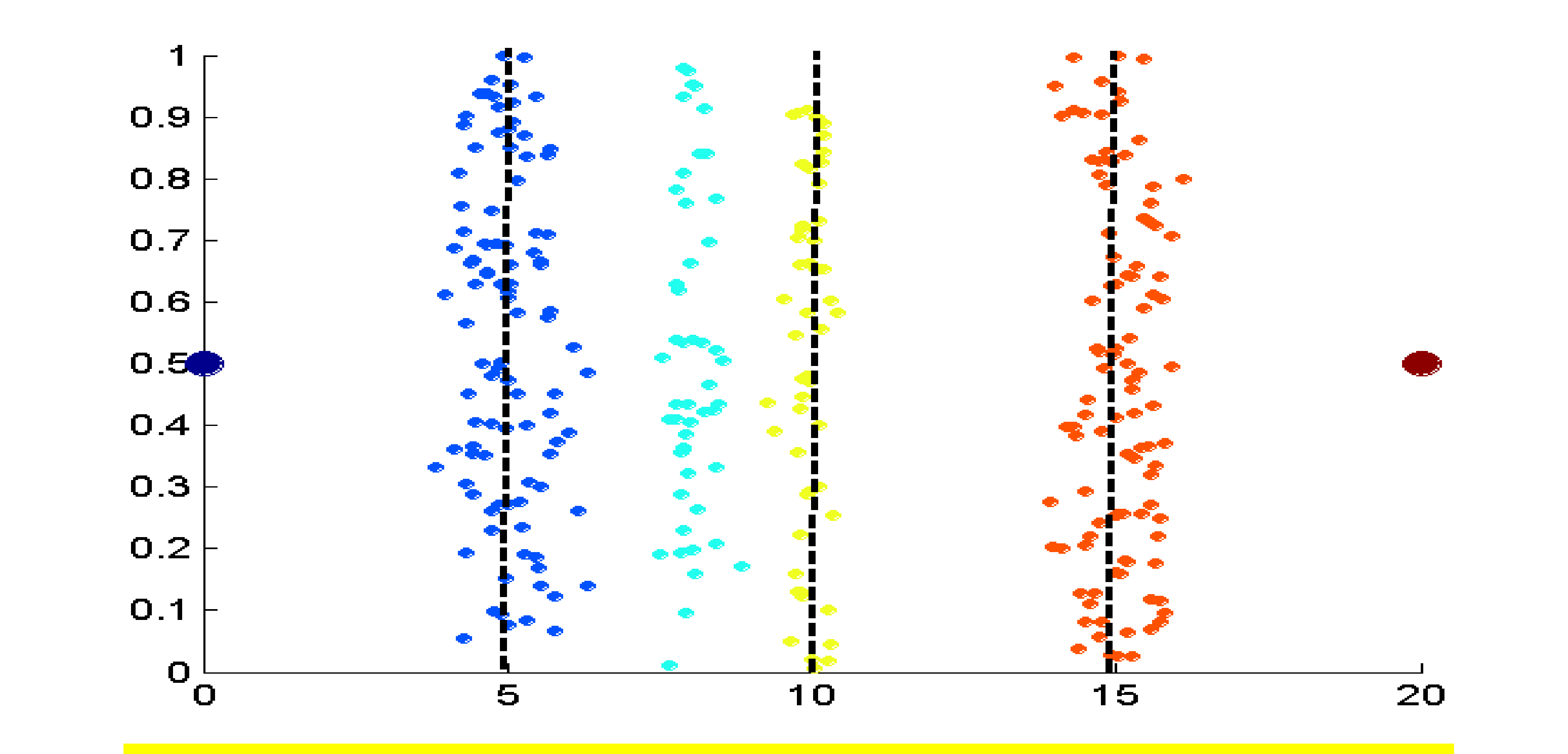
DATA TRANSFORMATION : DISCRETIZATION

- **Binning**: Top-down split, unsupervised
- **Histogram analysis**: Top-down split, unsupervised
- **Clustering analysis**: Unsupervised, top-down split or bottom-up merge
- **Decision-tree analysis**: Supervised, top-down split
- **Correlation analysis**: Unsupervised, bottom-up merge

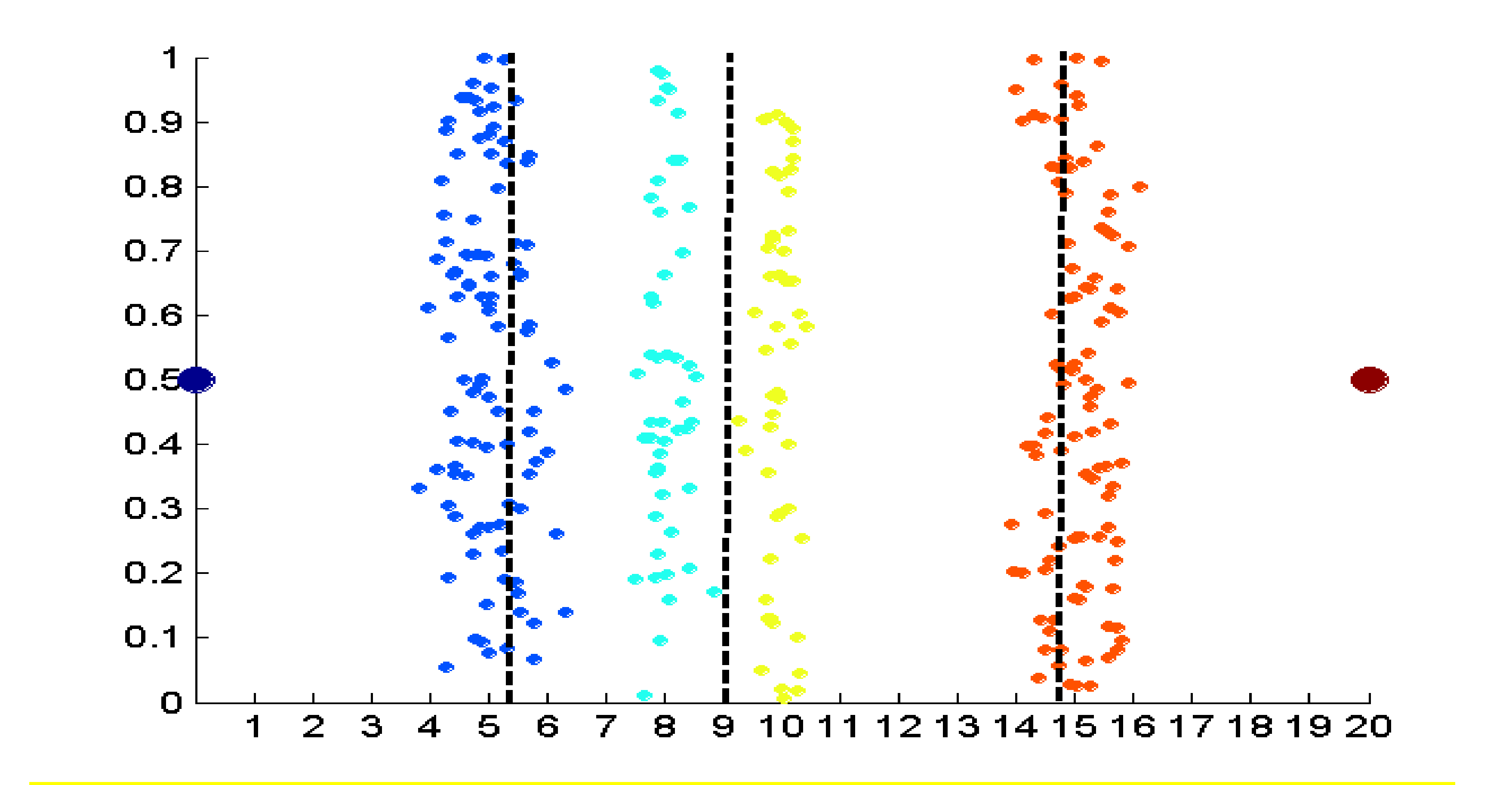
DISCRETIZATION WITHOUT SUPERVISION: BINNING VS. CLUSTERING



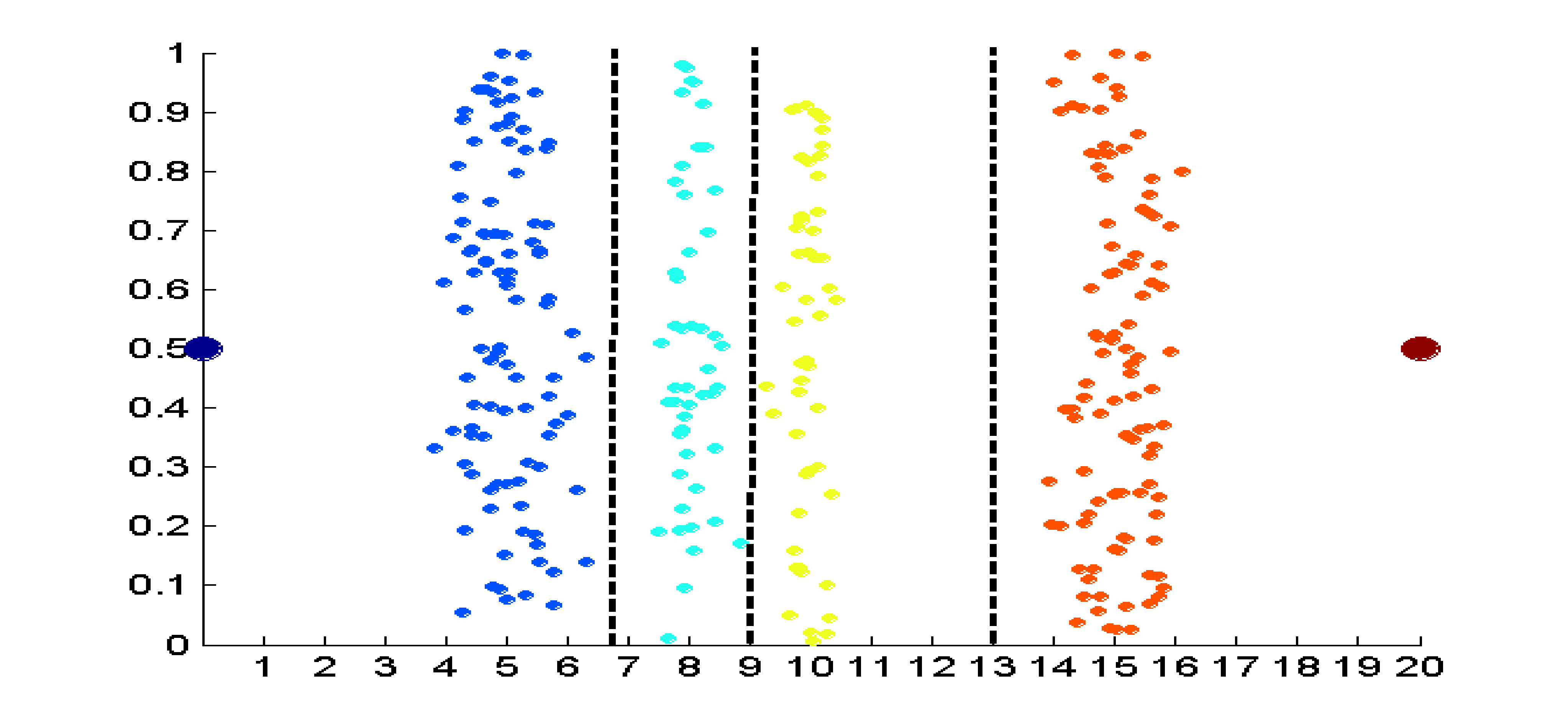
Data



Equal width (distance) binning



Equal depth (frequency) (binning)



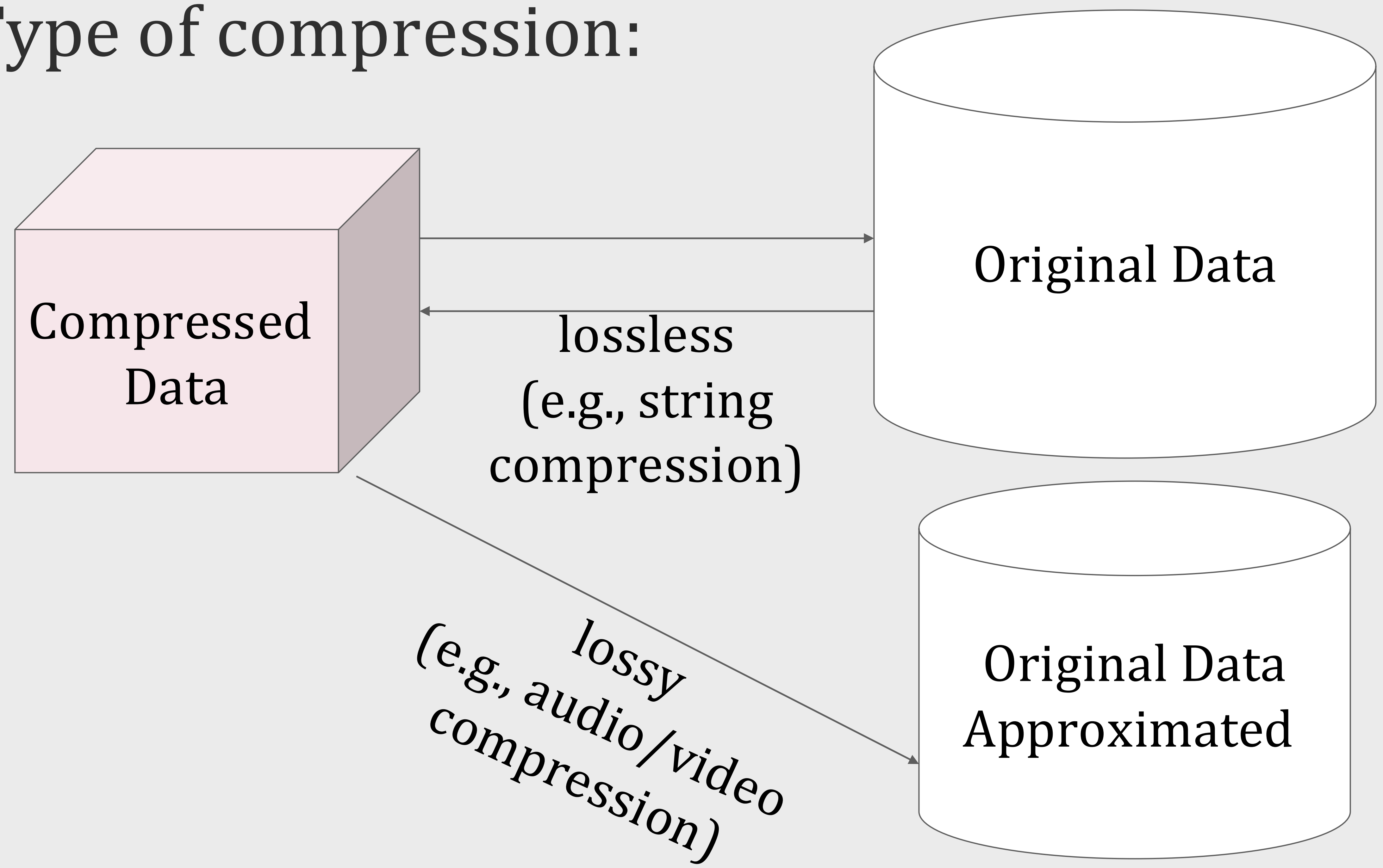
K-means clustering leads to better results

DISCRETIZATION BY CLASSIFICATION & CORRELATION ANALYSIS

- Classification (e.g., decision tree analysis)
  - Supervised: Given class labels, e.g., cancerous vs. benign
  - Using entropy to determine split point (discretization point)
  - Top-down, recursive split
  - Details to be covered in Chapter “Classification”
- Correlation analysis (e.g., Chi-merge: χ2-based discretization)
  - Supervised: use class information
  - Bottom-up merge: Find the best neighboring intervals (those having similar distributions of classes, i.e., low χ2 values) to merge
  - Merge performed recursively, until a predefined stopping condition

DATA REDUCTION : DATA COMPRESSION

- Data Compression: String Compression or Audio/Video Compression
- Type of compression:

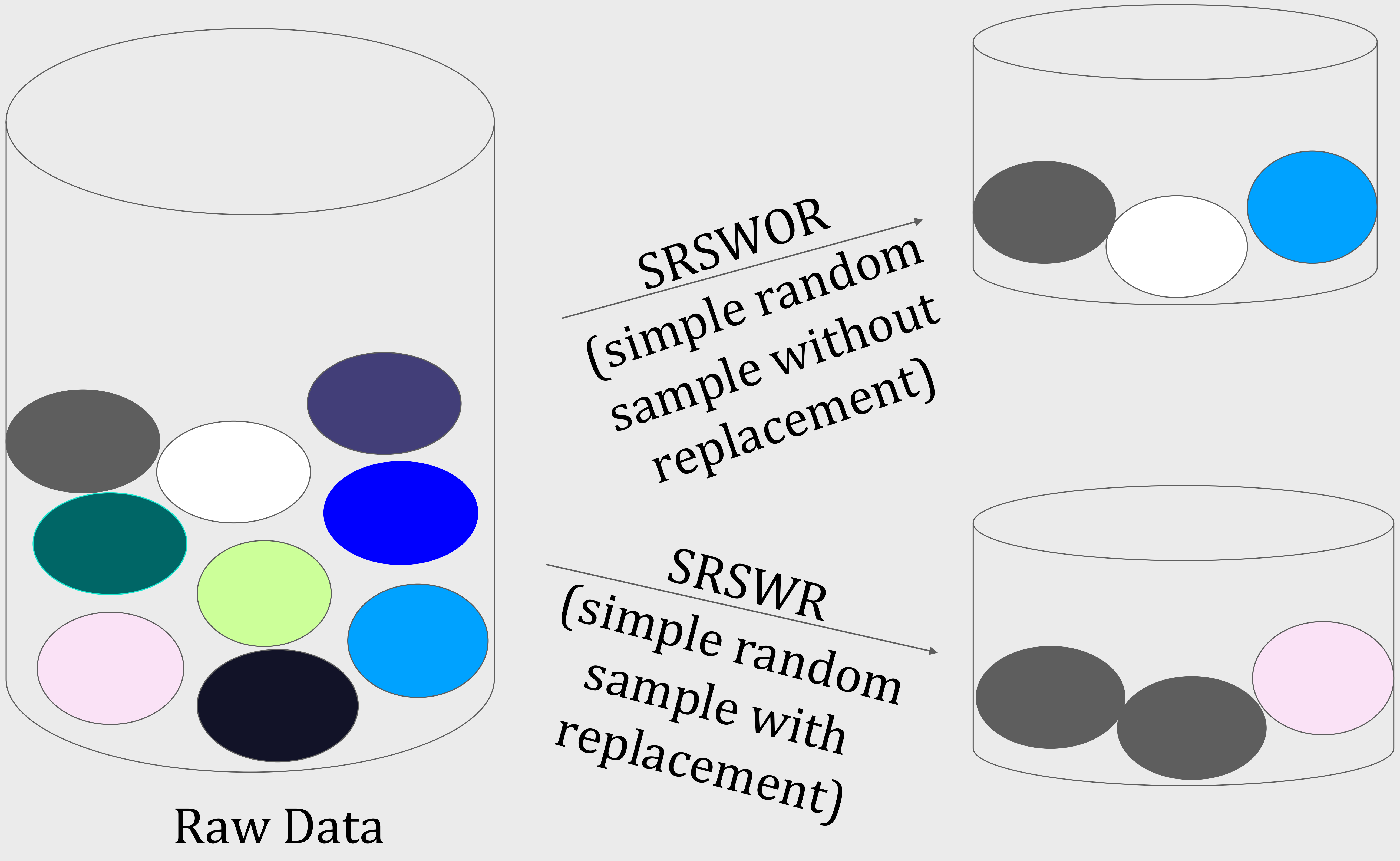


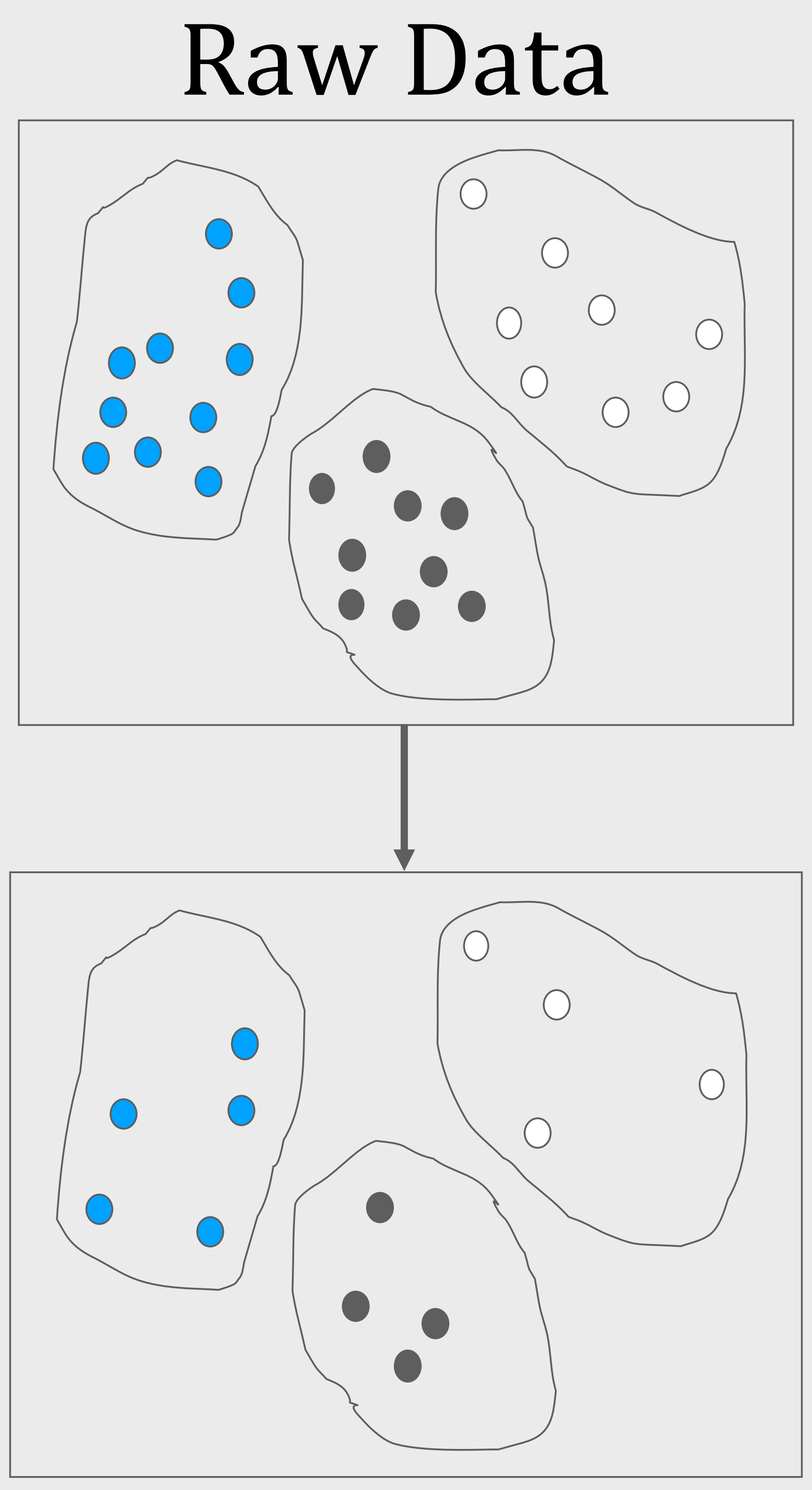
- Data reduction and dimensionality reduction may also be considered as forms of data compression

BINUS HIGHER EDUCATION

DATA REDUCTION : SAMPLING

- Sampling: obtaining a small sample *s* to represent the whole dataset *N*
- Key principle: Choose a representative subset of the data





Cluster/Stratified Sampling

TYPE OF SAMPLING

- **Simple random sampling**: equal probability of selecting any particular item
- **Sampling without replacement**
  - Once an object is selected, it is removed from the population
- **Sampling with replacement**
  - A selected object is not removed from the population
- **Stratified sampling**
  - Partition (or cluster) the data set, and draw samples from each partition (proportionally, i.e., approximately the same percentage of the data)

DATA REDUCTION

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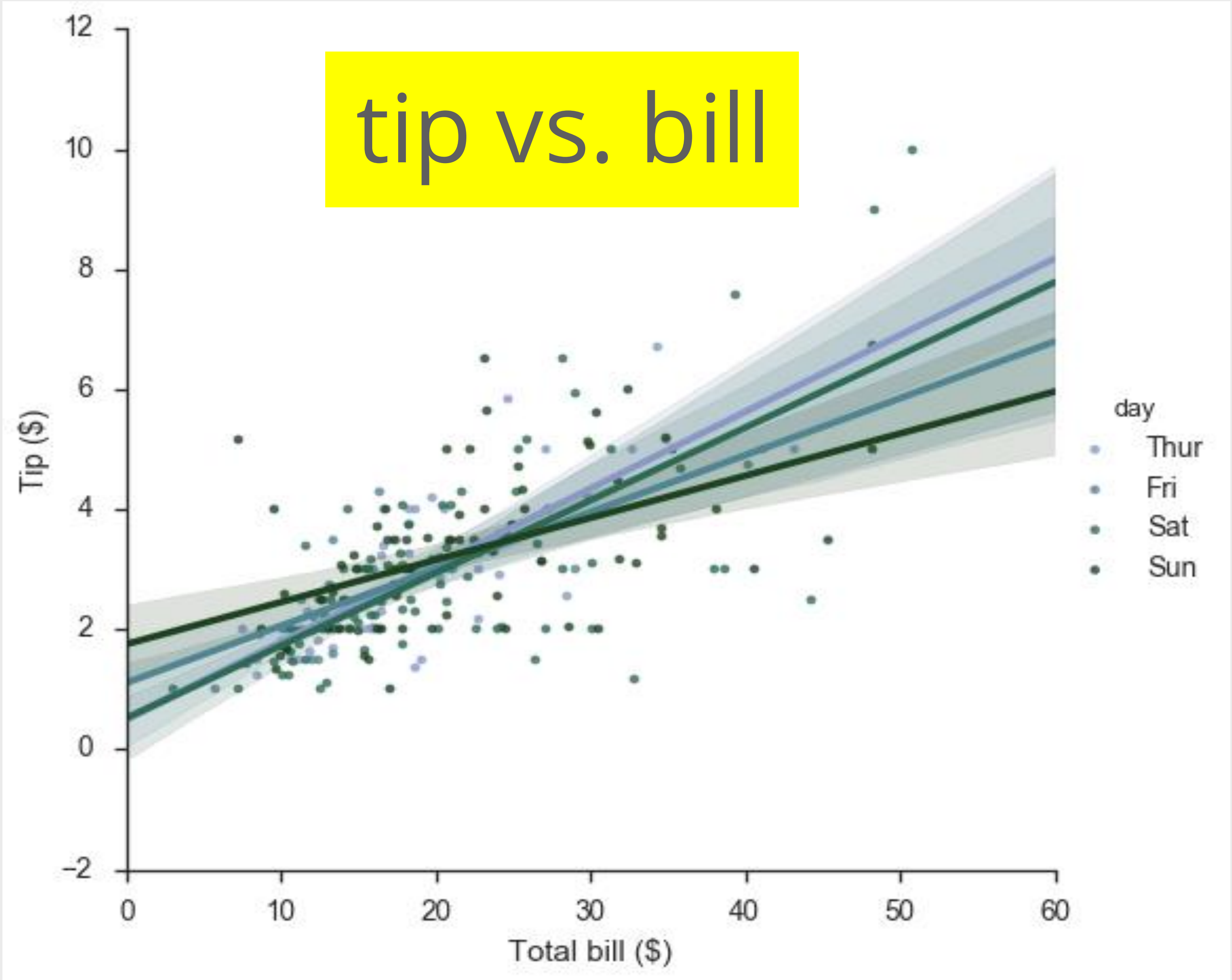
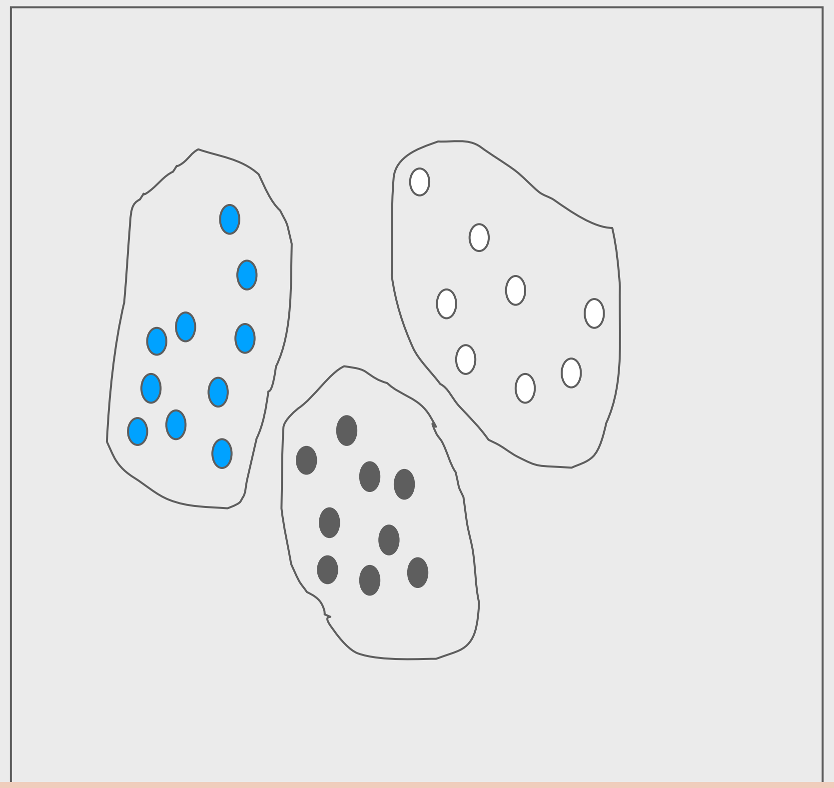
DATA REDUCTION

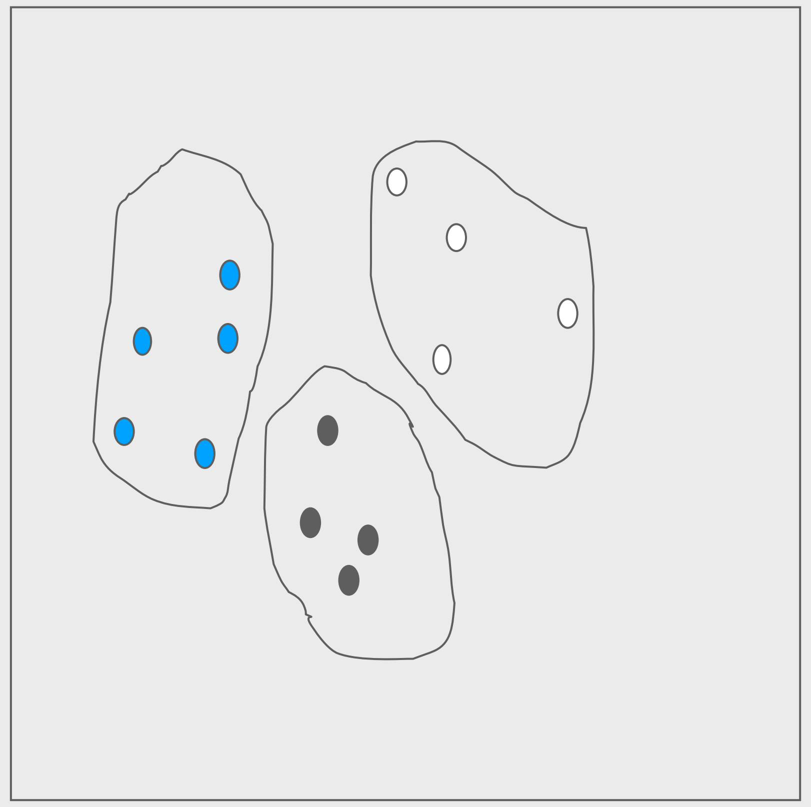
- To obtain a reduced representation of the dataset that is much smaller in volume but yet produces the same (or almost the same) analytical results
- Why data reduction?—A database/data warehouse may store terabytes of data
  - Complex analysis may take a very long time to run on the complete data set
- Methods for data reduction (also data size reduction or numerosity reduction)
  - Parametric: Regression and Log-Linear Models
  - Non-parametric: Histograms, clustering, sampling

DATA REDUCTION: PARAMETRIC VS. NON-PARAMETRIC METHODS

- Reduce data volume by choosing alternative, smaller forms of data representation
- Parametric methods (e.g., regression)
  - Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
  - Ex.: Log-linear models—obtain value at a point in m-D space as the product on appropriate marginal subspaces
- Non-parametric methods
  - Do not assume models
  - Major families: histograms, clustering, sampling, …




Clustering on the Raw Data


Stratified Sampling

DATA REDUCTION : DIMENSIONALITY

- Curse of dimensionality
  - When dimensionality increases, data becomes increasingly sparse
  - Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
  - The possible combinations of subspaces will grow exponentially
- Dimensionality Reduction: Reducing the number of random variables under consideration, via obtaining a set of principal variables
- Advantages of dimensionality reduction
  - Avoid the curse of dimensionality
  - Help eliminate irrelevant features and reduce noise
  - Reduce time and space required in data mining

DIMENSIONALITY REDUCTION METHODS

- Dimensionality reduction methodologies
  - Feature selection: Find a subset of the original variables (or features, attributes)
  - Feature extraction: Transform the data in the high-dimensional space to a space of fewer dimensions
- Some typical dimensionality reduction methods
  - Principal Component Analysis
  - Attribute Subset Selection
  - Nonlinear Dimensionality Reduction

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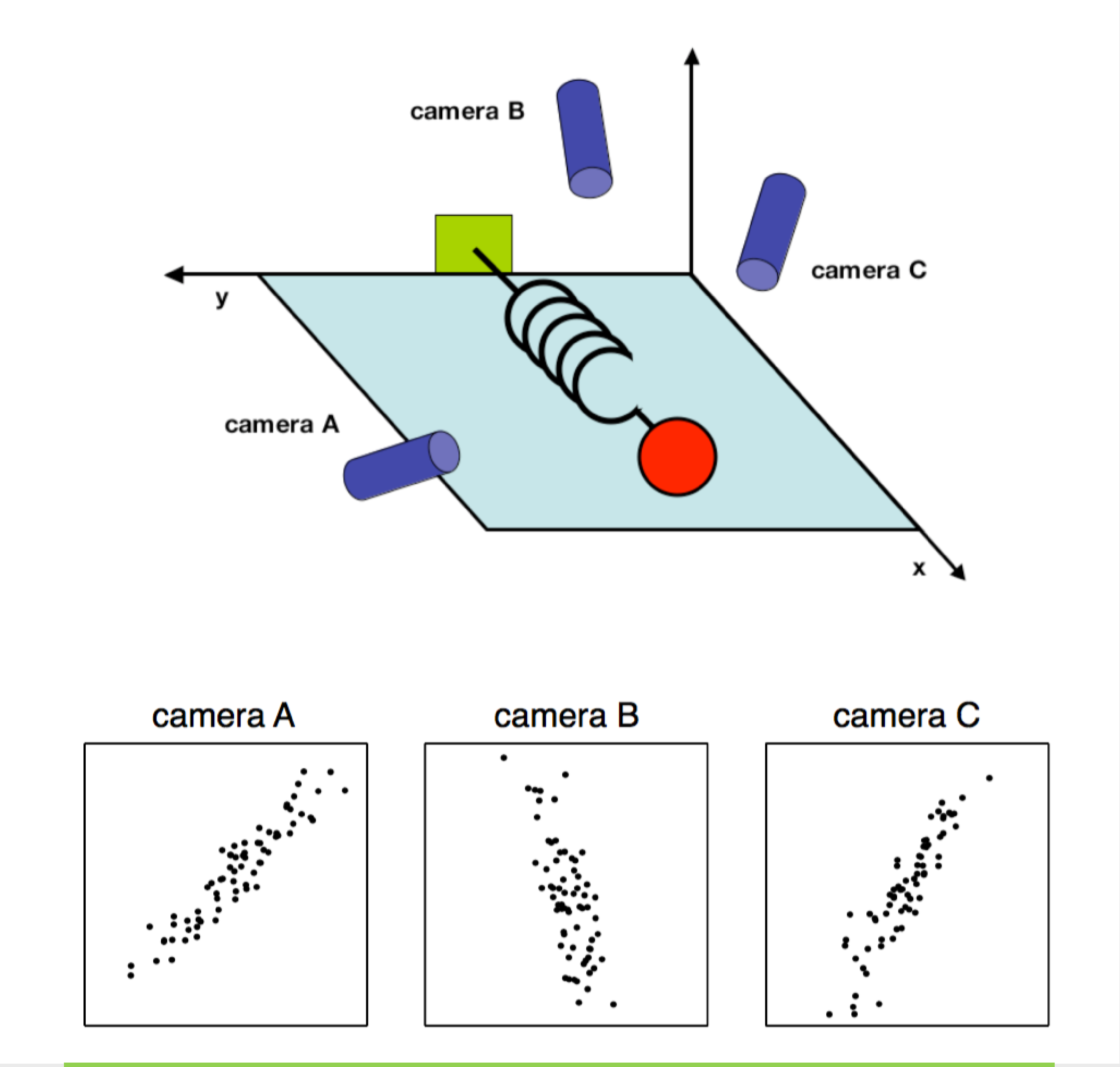
PRINCIPAL COMPONENT ANALYSIS (PCA)

- PCA: A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components
- The original data are projected onto a much smaller space, resulting in dimensionality reduction
- Method: Find the eigenvectors of the covariance matrix, and these eigenvectors define the new space
- Works for numeric data only



Ball travels in a straight line. Data from three cameras contain much redundancy

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PRINCIPAL COMPONENT ANALYSIS (PCA)

- Given N data vectors from n-dimensions, find $k \le n$ orthogonal vectors (principal components) best used to represent data
  - Normalize input data: Each attribute falls within the same range
  - Compute k orthonormal (unit) vectors, i.e., principal components
  - Each input data (vector) is a linear combination of the k principal component vectors
  - The principal components are sorted in order of decreasing “significance” or strength
  - Since the components are sorted, the size of the data can be reduced by eliminating the weak components, i.e., those with low variance (i.e., using the strongest principal components, to reconstruct a good approximation of the original data)

ATTRIBUTE SUBSET SELECTION

- Another way to reduce dimensionality of data
- Redundant attributes
  - Duplicate much or all of the information contained in one or more other attributes
  - E.g., purchase price of a product and the amount of sales tax paid
- Irrelevant attributes
  - Contain no information that is useful for the data mining task at hand
  - Ex. A student’s ID is often irrelevant to the task of predicting his/her GPA

ATTRIBUTE CREATION (FEATURE GENERATION)

- Create new attributes (features) that can capture the important information in a data set more effectively than the original ones
- Three general methodologies
  - Attribute extraction: Domain-specific
  - Mapping data to new space (see data reduction): E.g., Fourier transformation, wavelet transformation, manifold approaches (not covered)
  - Attribute construction: Combining features (see discriminative frequent patterns in Chapter on “Advanced Classification”), Data discretization

SUMMARY

- Data transformation: normalization, discretization, data compression and sampling
- Data reduction: Parametric (Regression and Log-Linear Models), Non-parametric (Histograms, clustering, sampling)
- Dimensionality Reduction: Reducing the number of random variables under consideration, via obtaining a set of principal variables

EXERCISE

1. Explain the purpose of data transformation in the KDD process. Why is it important before model training?
2. Suppose you have a dataset containing numerical values ranging from 0 to 10,000. Which transformation technique would you use to scale the values between 0 and 1?
3. Explain the concept of PCA (Principal Component Analysis) in data reduction. How does it help improve model efficiency?

REFERENCES

Han, J., Kamber, M., & Pei, J. (2023). Data Mining: Concepts and Techniques (4rd ed.). San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.